

Supporting Information for "Microfluidics and spectral induced polarization for direct observation and petrophysical modeling of calcite dissolution"

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Introduction This supporting information file provides additional material for the geochemical simulation with text presenting the Damköhler and Péclet numbers to support the hypothesis of reaction kinetics fast enough to reach thermodynamic equilibrium at each time step of the simulation. The file also provides additional information for the petrophysical modeling, which are the results of the microfluidic channel modeling with EIDORS and the graphs and maps of the residuals for the real and imaginary components of the modeled complex electrical conductivity. Finally, this file describes and gives access to movies and datasets obtained from image processing and geochemical simulation.

Text S1. In the following paragraph, we give supplementary information about the geochemical model. We present the values for Péclet and Damköhler numbers of our experiment. They are used to justify the hypothesis of thermodynamic equilibrium at each time step of the pseudo-transport simulation. The Damköhler and Péclet numbers are calculated according to the formulas of Soulaïne et al. (2017):

$$Da = \frac{k}{A_e D_A} \quad (1)$$

and

$$Pe = \frac{v_0 l}{D_A}, \quad (2)$$

where the dissolution kinetics k is $0.5 \text{ mol m}^{-2} \text{ s}^{-1}$ in the acid branch according to Palandri and Kharaka (2004) and the diffusion coefficient D_A is $5 \cdot 10^{-9} \text{ m}^2 \text{ s}^{-1}$ (Soulaïne et al., 2017).

It is assumed that the surface area A_e is between the value of the surface area of a cylinder ($5 \cdot 10^{-7} \text{ m}^2$) and a surface with defects ($5 \cdot 10^{-6} \text{ m}^2$). This gives a Damköhler Da number between 20 and 200.

Based on the dimensions of the channel (1.50 mm width and 0.15 mm height) and the flow rate value set at 1.25 mL h^{-1} ($0.35 \text{ mm}^3 \text{ s}^{-1}$), the velocity of the solution is 1.56 mm s^{-1} ($1.56 \cdot 10^{-3} \text{ m s}^{-1}$). The distance between the channel wall and the grain of 0.25 mm is considered as the characteristic length l . This gives a Péclet number $Pé$ of 78.

The Damköhler and Péclet numbers are large in front of 1. In the representation of the Péclet-Damköhler diagram ($Pé - Da$), we would be in a region where the reaction is rapid to diffusion and where the advective flow affects the dissolution pattern (blue zone in Figure 11 of Soulaire et al., 2017). This is consistent with the shape taken by the calcite grain during the reaction. We are therefore in a regime where the control of chemistry is not impacted by the kinetic limitation of calcite dissolution and can assume that equilibrium is reached in 10 s.

Figure S1. Using EIDORS (Electrical Impedance and Diffuse Optical tomography Reconstruction Software) a mesh of the geometry of the channel and the electrodes is generated (Adler & Lionheart, 2006). From this, a simulation of the streamlines is performed for the channel with homogeneous conductivity of 1 S m^{-1} , an infinite resistor for the calcite sample, and a current resistor $z_c=1 \Omega$ for the electrodes. The streamlines are straight and parallel between them. They go around the calcite sample and are only distorted at the location of the measuring electrodes. Lower current resistor values for the electrodes (0.1 and 0.01Ω) have been tested, showing similar profiles. This numerical

result indicates that the initial calcite has a relatively minor effect on the trajectory of the streamlines compared to the geometry of the confined channel. This explains setting a constant cementation exponent $m = 1$ (Glover, 2009).

Figure S2. The color scale on the maps of the residuals for each component σ' and σ'' highlights the curve of the CEC in dark blue, which minimizes the residuals for each component independently. For times earlier than $t = 2.45$ h, the curves are similar for both σ' and σ'' . For later times, only the residuals of component σ' are minimized with a CEC $> 0 \text{ mEq g}^{-1}$ and after the end of the dissolution ($t = 4.29$ h), the residuals are independent of the CEC. The discrepancy between the components highlights that at this frequency the CEC, related to the specific surface area, impacts more conduction than polarization due to the morphological changes caused by calcite dissolution which are more important at the beginning of the experiment as observed on the microscope images in the study from Rembert, Stolz, Soulaine, and Roman (2023).

The curves on Fig. S2c correspond to the residuals for each component σ' and σ'' when using the CEC value that minimizes the residuals on both components following Eq. (10). The residuals are consistently of a low order of magnitude compared to the components to which they refer, indicating the effective adjustment of the model for the entirety of the experiment.

Data Set S1. Porosity, water saturation, and calcite sample perimeter from image processing.

Data Set S2. Porosity, water conductivity, and pH from geochemical simulation.

Data Set S3. Real and imaginary components of the complex electrical conductivity at 2.5 Hz and CEC from petrophysical modeling.

Movie S1. Dissolution of the calcite sample with the detected contour superimposed in white on the grayscale images. Time, length scale, and flow direction are indicated. In case of problems launching the file, we recommend using VLC Media Player software.

Movie S2. Segmented images of the CO₂ bubbles produced by the calcite dissolution. Time, length scale, and flow direction are indicated. In case of problems launching the file, we recommend using VLC Media Player software.

References

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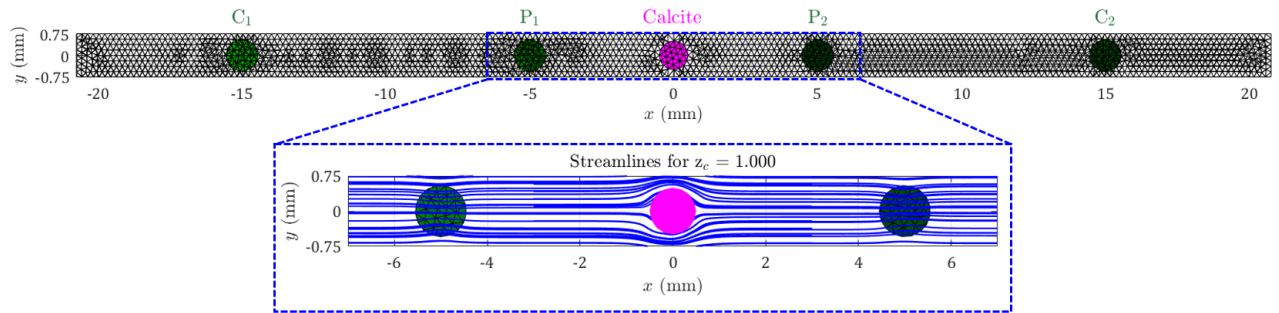


Figure S1. Mesh of the channel with the electrodes and the calcite sample. The view in the blue dashed box shows the simulated streamlines.

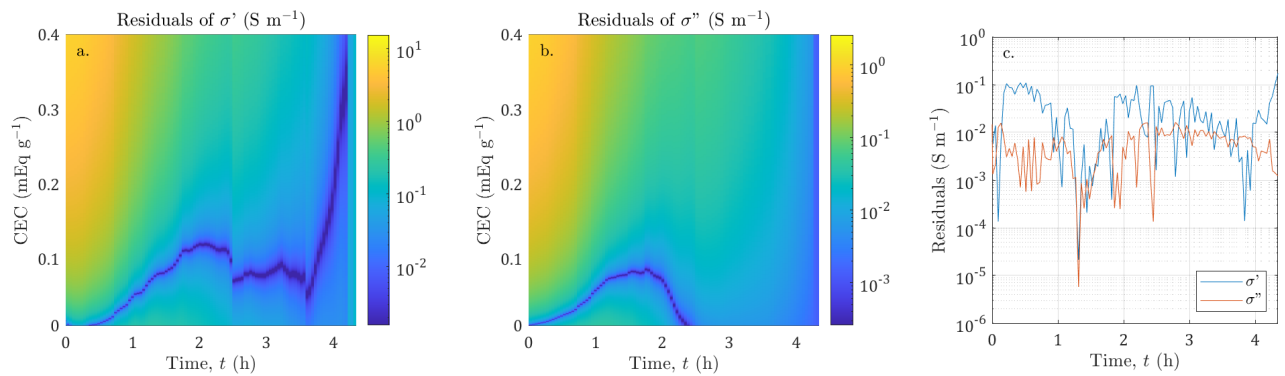


Figure S2. Maps of the residuals of the (a) real and (b) imaginary components of the complex electrical conductivity for all of the CEC values over time. (c) The residuals for the fitted CEC curve.